

Speech Recognition

Exercises about Hidden Markov Models (HMM)

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Exercise 01

Consider the following HMM:

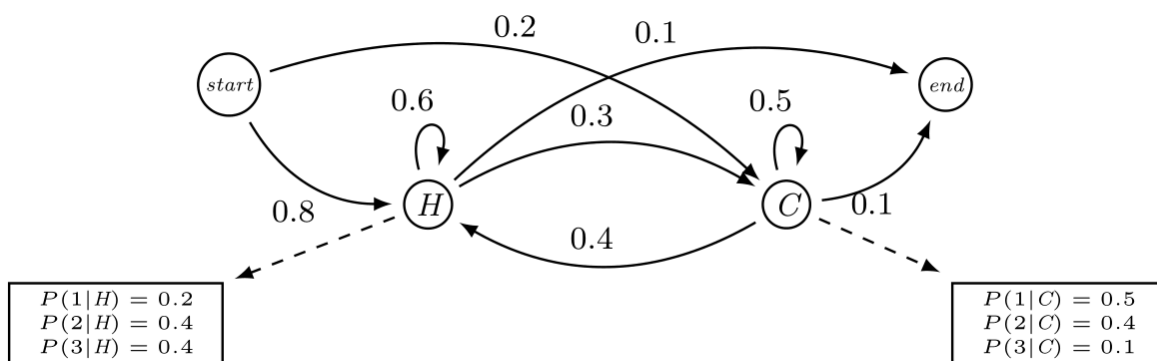


Figure 1: Hidden Markov Model HMM for inferring weather (hot = H, cold = C) from numbers of ice creams eaten by a person.

Part a)

What is the probability of traversing a path

$start \rightarrow C \rightarrow C \rightarrow end$

according to this HMM?

Part b)

What is the probability that, when observing the weather on three different subsequent days, of encountering at least two subsequent cold days, i.e., twice the event C, one following the other?

Part c)

What is the probability of an output 1 3 given this model?

Exercise 02

Suppose you have the following tagged corpus (each word followed by its POS Tag, in brackets, adapted from: RRG Baumbank, based on George Orwell's "1984", <https://rrgparbank.phil.hhu.de/>)

sent 1: (it PRO) (was AUX) (the OP) (telescreen N) (. .)

sent 2: (the OP) (Thought N) (Police N) (used V) (the OP) (telescreen N) (. .)

sent 3: (the OP) (Thought N) (Police N) (mattered V) (for P) (them PRO) (. .)

sent 4: (Winston N) (kept V) (his OP) (back N) (turned V) (to P) (the OP) (telescreen N) (. .)

sent 5: (Winston N) (wanted V) (to CLM) (turn V) (. .)

(. .) is the end symbol of a sentence.

Part a)

State the transition probabilities of the HMM that results from this corpus by counting occurrences, i.e. as Maximum Likelihood Estimation (MLE). It is sufficient if you draw the corresponding graph.

Part b)

Also, give all emission probabilities that contain the words "Winston", "mattered", "to", "them" and "." concern, and which are not 0.

Part c)

Using this model, what is the best tag sequence for the sentence "Winston mattered to them ."? To answer this question, calculate the Viterbi matrix, which deals with the model and this input results. It suffices to specify the entries in the matrix that are $< > 0$.